



MAJI METRICS

Predictive Analytics for Reliable Water Systems

OUR TEAM



**Lead
Statistician**

Abdullahi Abdi
Hassan



**Data
Architect**

Claire



**Feature
Engineer**

Yvonne



**ML
Engineer**

Dahir



**Visualization
Specialist**

Lauren



**Pipeline
Engineer**

Samantha

WHAT WE DO



Strengthen
water infrastructure
using data



Support
SDG 6: Clean Water
and Sanitation



Improve access
across rural
Kenya

KEY Pillars

1



**Predictive
Modeling**

Identify failures
before they occur

2



**Smart
Decision-Making**

Guide efficient
maintenance and
resource use

3



**Inclusive
Access**

Prioritize high-risk
and underserved
areas

4



**Sustainable
Outcomes**

Enhance long-term
system reliability



Better Insights. Smarter Actions. Stronger Communities.

The Human Problem (Hook)



The Reality for Communities Across Kenya



1

Dry Taps, Every Day

Every day, thousands of Kenyans wake up to a dry tap.



2

Broken When It Matters Most

30% of rural water points in sub-Saharan Africa are broken at any given time.



3

Miles for Water That Should Flow

Communities **walk miles** for water that should flow from their local pump.



4

Repairs Only After Failure

Repairs only happen after failure → **days or weeks without water.**



5

Millions Invested. Neglect Prevails.

Donors invest millions in new pumps, but existing ones **crumble from neglect.**



The question isn't "how do we build more?"
It's "how do we *keep what we have* **WORKING?**"



Our Solution — Predictive Maintenance



We built an AI that predicts which water pumps will fail **BEFORE** they break.



TRADITIONAL:

Pump breaks



Community suffers



Repair



(Days/weeks without water)



MAJI METRICS:

AI flags risk



Maintenance arrives



Pump stays functional



(Zero disruption)



Think of it like a “**check engine light**”
for every water point in Kenya.



The Data — What We Analyzed



Turning Kenya's Water Data Into Predictive Power



Source:

Kenya Water Points
Dataset (WPdx)



Scope:

21,953 water points
across all 47 counties

Key Information Per Water Point:



Location
(GPS
coordinates)



**Technology
type**
(hand pump,
motorized,
tapstand)



**Installation
year & age**



**Population
served**



**Functional
status**
(Functional /
Broken)



**Management
type**
(Community,
Government,
Private)



**This is the largest, most comprehensive
water infrastructure dataset in Kenya.**

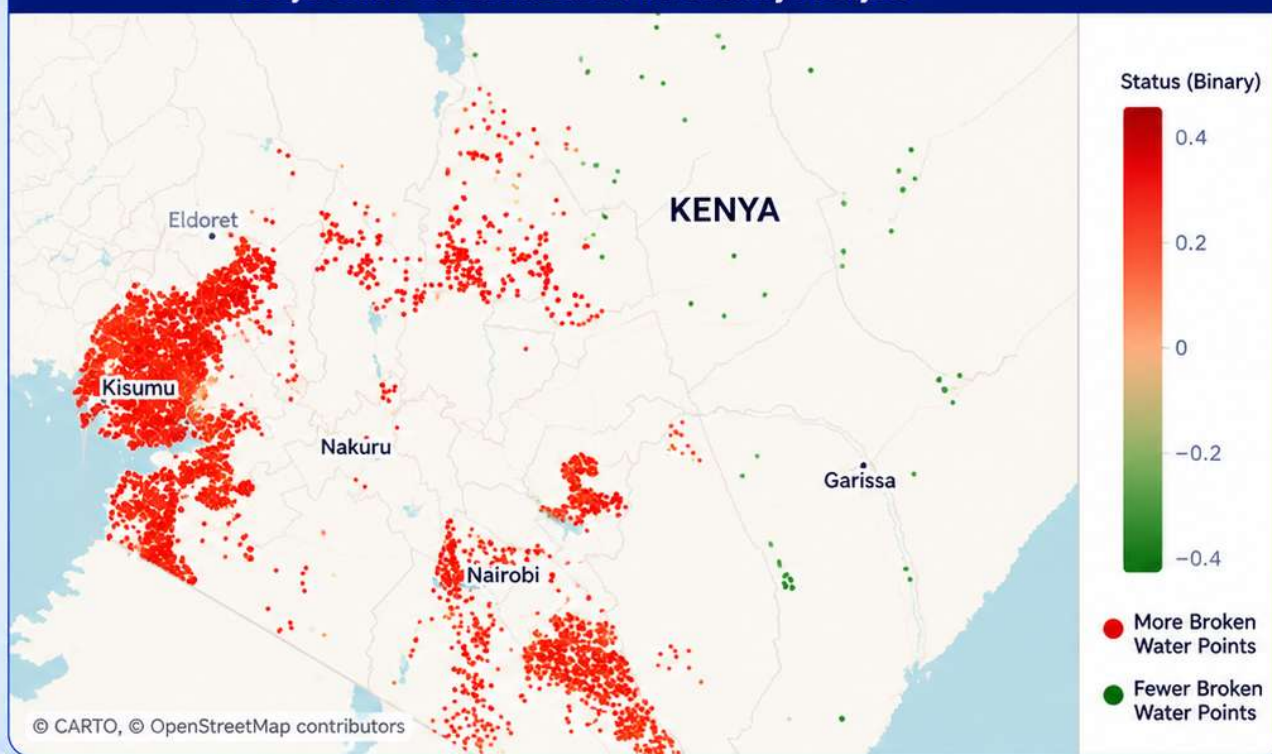


What We Discovered — Visual Insights



Mapping Broken Water Points Across Kenya

Kenya Water Infrastructure: Full Country Analysis



KEY FINDINGS



Western Kenya shows **dense clusters** of broken pumps.



Nairobi outskirts also **high-risk**.



Northern & Eastern regions have **fewer failures** (but also fewer pumps).

IMPLICATION



Regional factors matter.
One-size-fits-all maintenance doesn't work.



Data shows us **WHERE** the problems are.
AI helps us **PREDICT** which ones will fail next.

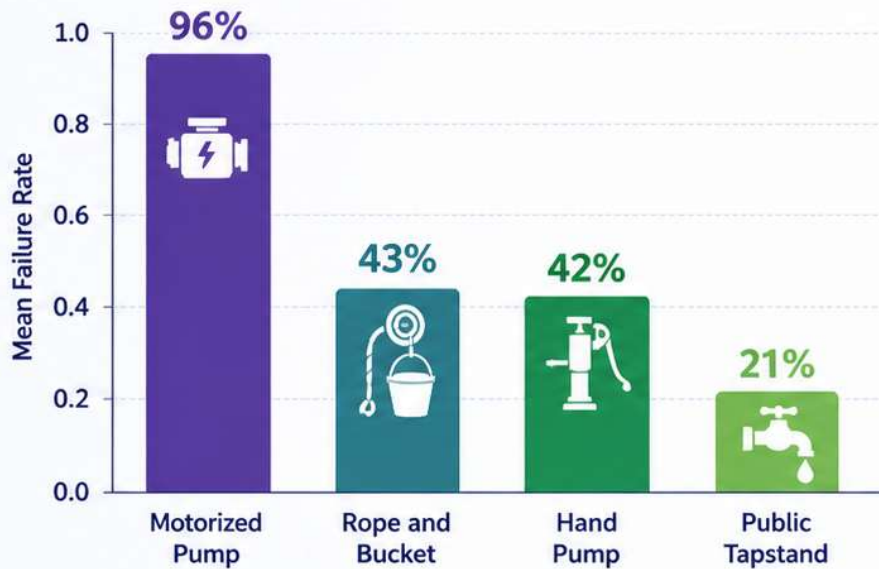


Technology Matters



Failure Rates by Pump Type

Failure Rate by Technology Category



Data Source: Kenya Water Points Dataset (WPdx)
Scope: 21,953 water points across all 47 counties

Technology

Failure Rate

Action



Motorized
Pumps

Near
100%



PHASE OUT



Rope &
Bucket

~40%



INSPECT



Hand
Pumps

~40%



MONITOR



Public
Tapstands

Lowest



RESILIENT



Takeaway: What we install matters as much as where we install it.

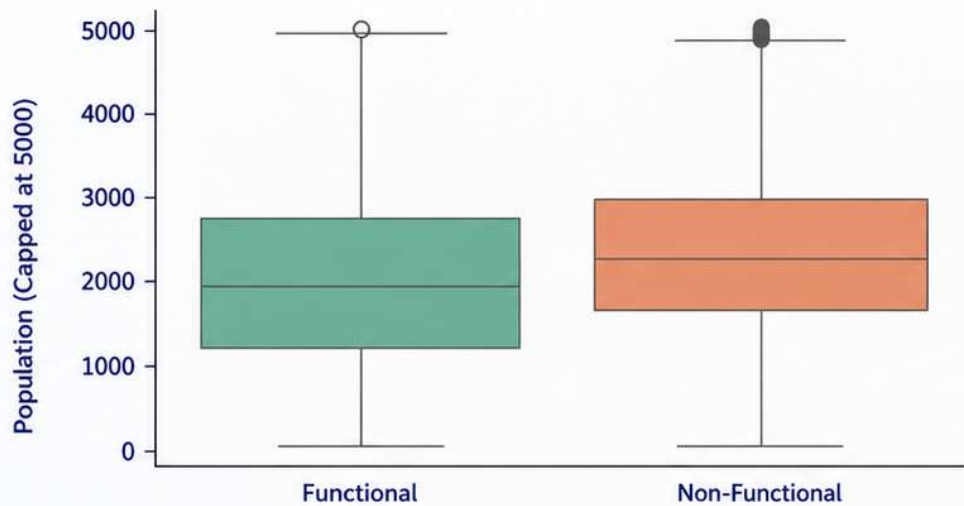


High-Demand = High Failure



Population Density vs. Pump Status

Local Population vs. Water Point Status



High population areas are significantly more likely to experience pump failure.

The Insight



Pumps serving more people break more often



Heavy use = faster wear and tear



High-density communities need **MORE** resilient infrastructure **OR** more frequent maintenance.



Takeaway: Plan smarter, maintain regularly, and invest where demand is highest to keep communities flowing.



How We Built the Prediction Model



AI Models Trained. Performance Compared.

We trained 5 different AI models and compared them:

Model	What It Does
Logistic Regression	Simple baseline
Random Forest	Ensemble of decision trees
Extra Trees	Fast random forest variant
XGBoost	Extreme Gradient Boosting
CatBoost	CHAMPION — Handles categories best

How We Trained and Validated the Models



80%
Training
Data



20%
Testing
Data

5-Fold Random Split Validation

We validated across 5 different random splits to ensure reliability.



This ensures our model's performance is robust, consistent, and generalizes well to unseen data.



Takeaway: We used the best tools, the right process, and rigorous validation to build a model you can trust.

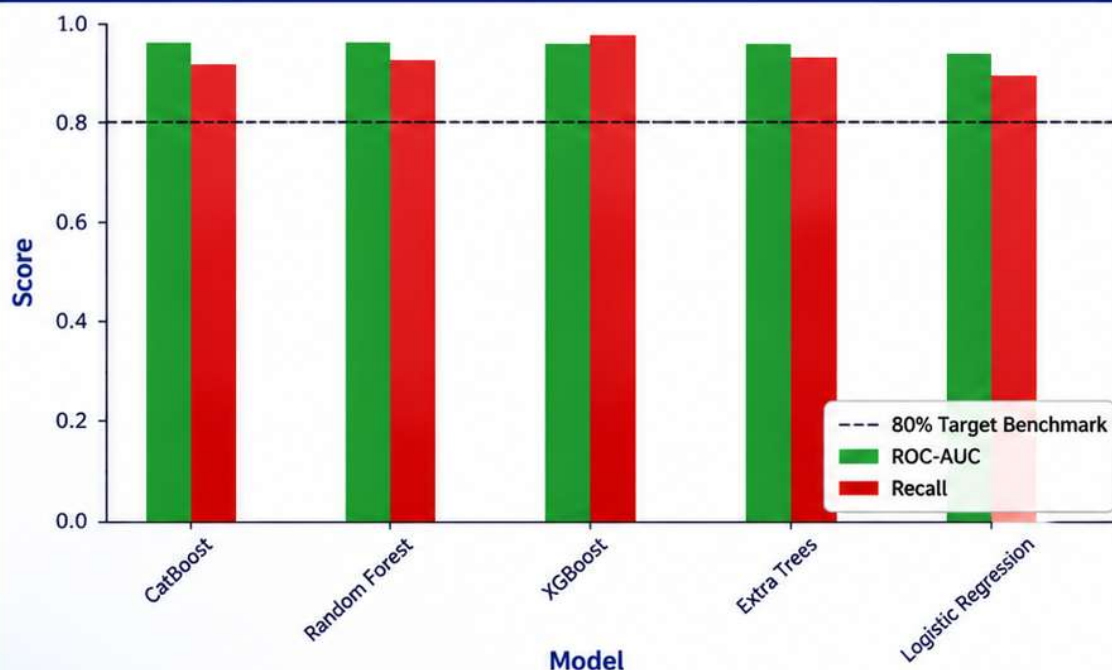


The Results — CatBoost Wins



Model Performance Tournament

Model Performance Comparison (Threshold = 0.2)



CatBoost Performance Summary

Metric	CatBoost Score	Target	Status
ROC-AUC	96.9%	≥80%	✓ EXCEEDS
Recall (Broken)	89.2%	≥80%	✓ EXCEEDS
Precision	90.4%	—	✓ HIGH

Why CatBoost is Champion

- ✓ Highest overall accuracy
- ✓ Most stable across different data splits
- ✓ Best balance of finding broken pumps AND avoiding false alarms.



Takeaway: CatBoost delivers the best performance, balance, and reliability for predicting pump failures.



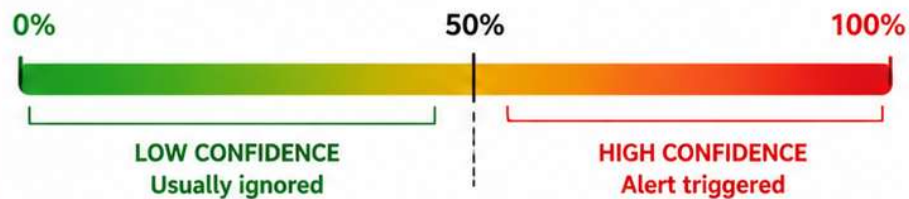
Our Secret Weapon — The 0.2 Threshold



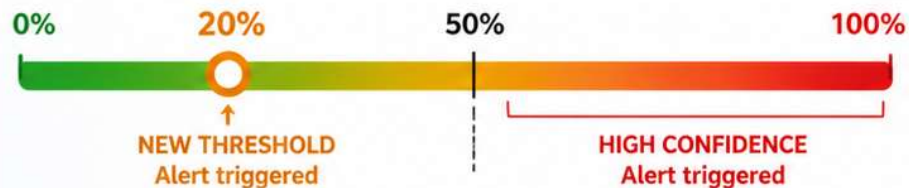
Lower Threshold. Earlier Alerts. Stronger Communities.

How AI Confidence Thresholds Work

Most AI models flag a pump as “broken” only if confidence > 50%



We Lowered the Bar to 20% Confidence



Standard:

“I’m 90% sure this pump will fail” → **Alert**



Maji:

“I’m 20% sure this pump MIGHT fail” → **Alert**

Why We Chose 0.2



Missing a broken pump
= a community suffers
for weeks



A false alarm
= one unnecessary
maintenance visit
(inexpensive)



We’d rather be
overly cautious
than miss a single
failing pump.



Takeaway: Lowering the threshold to 0.2 helps us catch potential failures earlier — protecting communities before problems begin.

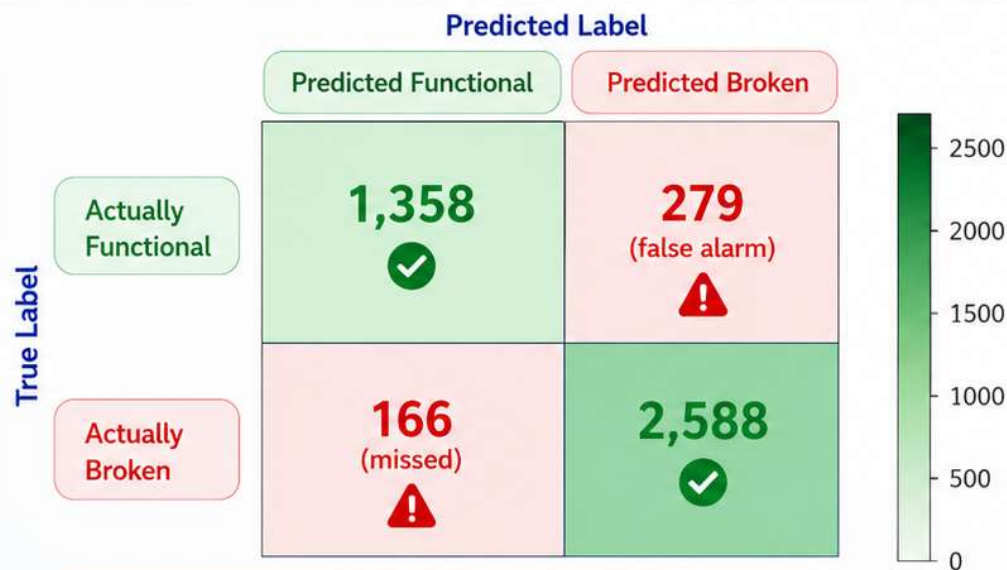


Confusion Matrix — Real Impact



CatBoost Model (Tuned for High Recall)

Final Confusion Matrix – CatBoost (Tuned for High Recall)



On ~ 4,400 test water points

What This Means



Broken pumps **CORRECTLY** identified
→ communities saved

2,588 ✓



Broken pumps **MISSED**
→ we want this closer to zero

166 ⚠



False alarms
→ minor operational cost, acceptable trade-off

279 ⚠



We prioritize **finding broken pumps**
to prevent communities from going without water.



Takeaway: CatBoost is tuned to catch as many broken pumps as possible — because the **cost of missing one** is far greater.



Our Final Recommendations



To the Ministry of Water & County Governments:

Priority	Action	Timeline
1	 Deploy CatBoost model as national early warning system	 Immediate
2	 Target maintenance to Western Kenya & Nairobi hotspots	 Q1
3	 Flag Motorized Pumps for phase-out or intensified inspection	 Q1-Q2
4	 Install resilient Tapstands in high-population areas	 Q2-Q4



These actions will help prioritize resources, reduce failures, and improve water access across Kenya.



To NGO Partners
(Evidence Action, UNICEF):



Use our risk scores to prioritize field visits



Focus on water points with >20% predicted failure probability.



The Bottom Line



Maji Metrics transforms water infrastructure management from **REACTIVE** to **PREDICTIVE**.

Before:



Break



Suffer



Repair

(Days/weeks without water)

After:



Risk identified



Prevent



Continuous water access



We predict

89%

of failures before they happen.



We protect communities from “water poverty.”



We help Kenya achieve SDG 6 — Clean Water and Sanitation for all.

Better Data. Smarter Decisions. Reliable Water for Every Community.





MAJI METRICS

Data Science for Smarter Water Decisions



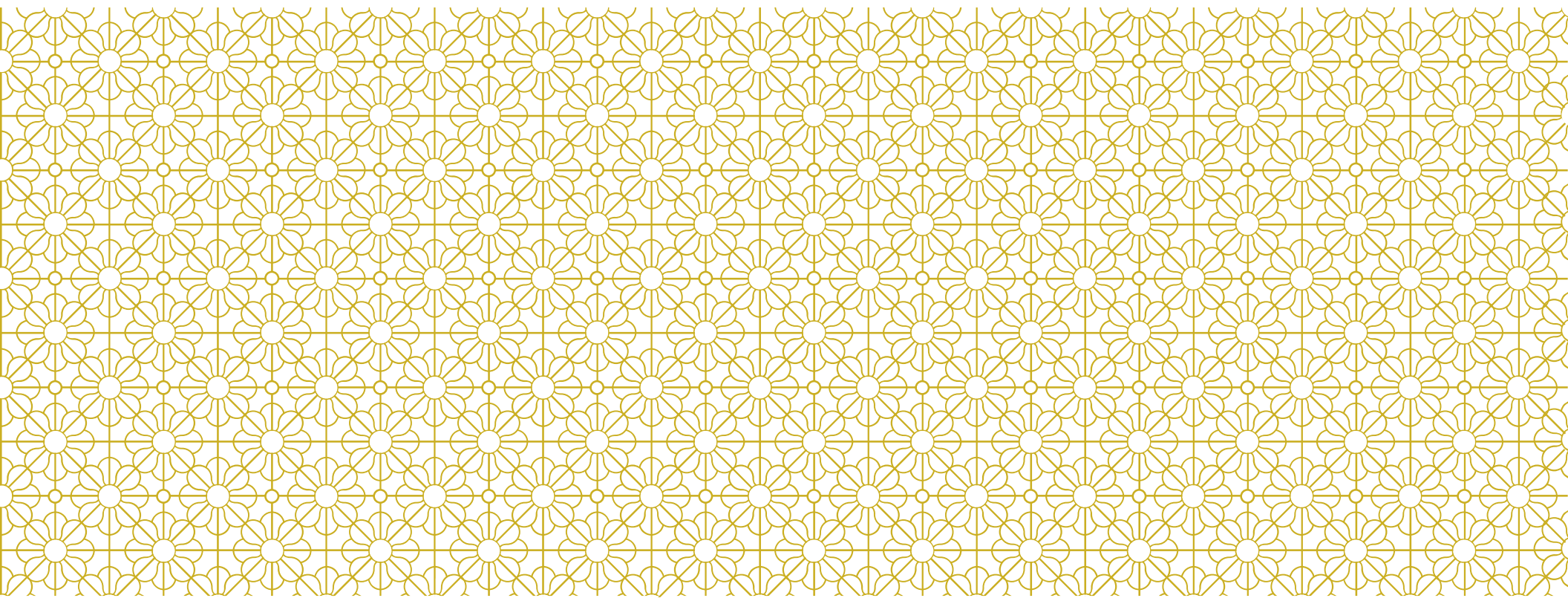
Thank You

Questions?



Better Data. Smarter Decisions.
Stronger Communities.





OUR LINK

<https://predictive-maintenance-equitable-water-infrastructure-group-2.streamlit.app/>